

ERIC ZHONG

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EDUCATION

University of Southern California

GPA: 4.00

M.S. Electrical Engineering (Computer Architecture and Chip Design Specialization)

Graduated May 2025

B.S. Computer Engineering and Computer Science

(B.S. + M.S.)

Honors: Top of Class, Presidential Merit Scholarship, Provost Research Fellowship, Engineering Honors Fellowship

Relevant Coursework

CS: Algorithms and Computing Theory, Operating Systems, Compilers, Networking, C++, Data Structures and OOP, Software Engineering

ECE: ASIC Design and Tapeout, Computer Microarchitecture, Computer Systems Architecture, VLSI Circuit Design, System-on-Chip (SoCs), FPGAs, Digital System Design, Network Processor Design, Parallel Computing Systems, Embedded and Distributed Systems

Math: Probability and Statistics, Linear Algebra, Discrete Math, Multivariable Calculus, Differential Equations

EXPERIENCE

NVIDIA

Santa Clara, CA

GPU Architect

June 2025—Present

- Driving hardware design, arch specification, **C++** simulation, and functional verification for NVIDIA's next-generation Feynman GPUs. Owned IPs include host **DMA engines** and work schedulers, **PCIe/CXL** controller endpoints, and interrupt controllers
- Writing custom firmware in **C** and **RISC-V assembly** for on-chip dynamic CUDA work scheduling and dependency resolution—enabling low-latency GPU work launch with reduced overhead and enhanced QoS
- Developing simulation libraries in **C++** and **C** within NVIDIA's proprietary resman **kernel driver** and MODS testing framework for initial architectural studies and bring-up of new GPU hardware + software features
- Performing functional and performance testing using a **Python DSL** across **C++** simulation, **RTL**, **FPGA** emulation, and silicon
- Creating a new **Python** and **Perl** based flow for communicating multi-die hardware device information to driver software
- Training and tuning a new **AI agent** for **DFD/DFT**—enabling autonomous debug and RCA of functional failures on end silicon

GPU Architecture Intern

May 2024—August 2024

- Developed a **C++** functional model (fmodel) for a new GPU hardware IP enabling low-latency, real-time interrupt message processing—facilitating RTL, driver, and CUDA development for NVIDIA's flagship Rubin GPUs
- Built and deployed a new lightweight unit testing infrastructure using **C++** and **gtest** within NVIDIA's proprietary simulation environment—accelerating functional and performance testing by up to a 20x speedup over existing fullchip testing workflows
- Refactored legacy **C++/SystemC** fmodel with **C++20** source to improve core sim performance and future code maintainability

USC Photonic Computing Lab (Opticore)

Los Angeles, CA

Research Fellow

August 2022—May 2024

- Co-invented and published a novel teraFLOP optical tensor processor with a 25 fJ/Op energy efficiency—offering up to a 40x efficiency improvement over current state-of-the-art electronic processors for machine learning workflows ([Publication Link](#))
- Designed and verified a high-throughput (10+ GSPS), low-latency (< 100 ns) **FPGA**-based waveform generation and signal processing module using **SystemVerilog** and **C++**. Synthesized design onto an AMD Zynq UltraScale+ RFSoc using **Vivado**
- Developed a custom **Python** library for host interfacing with lab **FPGAs** along **100G Ethernet** to facilitate optical experiments

IQVIA

Philadelphia, PA

Software Engineering Intern

May 2022—August 2022

- Engineered a high-throughput data pipeline using **Scala** and **Python** to process incoming market and research data feeds for 300+ healthcare products—aggregating and consolidating 40+ independent data sources for uniform ingestion and analysis
- Prototyped several multivariate regression models in **Python** using **Jupyter Notebook** for market forecasting across 20+ sectors—achieving an aggregate performance of <5% SMAPE across a 3-month production testing window
- Built and deployed a **CI/CD** pipeline for **Scala microservices** backed by **Oracle DB**—automating pre-prod testing and SQA

PROJECTS

NetFPGA SmartNIC

- Designed a high-throughput network switch capable of intelligent packet routing and preprocessing by payload content
- Architected and developed **Verilog** RTL for a multi-core NPU, custom network routing logic, and a software-programmable memory interface—achieving 2.5M packet inferences per second with **Gigabit Ethernet** processing on a **NetFPGA-1G**

Cryptographic Accelerator ASIC

- Developed architecture and **SystemVerilog** RTL for a power and area-optimized ASIC for efficiently computing variable delay functions (VDFs)—achieving 30+ million 256-bit hash computations per second with a module size of 105 um x 105 um
- Taped out design using Intel's proprietary 16nm FinFET process in February 2025 with post-silicon ASIC bring-up in Fall 2025

Multithreaded ARM Processor

- Designed a 32-bit, 4-way multithreaded, 8-stage pipelined processor in **Verilog** running a custom ARMv4-compatible ISA
- Synthesized and implemented RTL onto a Xilinx Virtex-II **FPGA** using **Xilinx ISE**, achieving the max 200 MHz oscillator clock
- Built a custom Clang-based compiler toolchain and backend for converting **C** code into hardware-executable **assembly**

SKILLS

Languages: C++, C, SystemVerilog, Verilog, Python, SystemC, Assembly, VHDL, Bash, Perl, Tcl

Tools/Techs: FPGAs, Vivado, Vitis, Verilator, cocotb, ModelSim, Icarus Verilog, Quartus, Xilinx ISE, CUDA, LLVM, Linux, Git, P4 Helix

Domains: Hardware + Software Co-design, Computer Architecture, Systems Software, Open-Source Silicon, GPUs